

Owain Thorp

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EDUCATION

Imperial College London

MSc Artificial Intelligence

London, UK

Sep 2025 – Present

- Provisional average: 90%
- Relevant Modules: Machine Learning, Mathematics for Machine Learning, Reinforcement Learning, Deep Graph-Based Learning, Probabilistic Inference, Robot Learning

University of St Andrews

MPhys Mathematics and Theoretical Physics

St Andrews, UK

Sep 2021 – Jul 2025

- Graduated with an 18.6/20.0 GPA (top 1%)

EXPERIENCE

Research Intern

Stealth Robot-Learning Startup (10-person team)

May 2026 – Present

London, UK

- Investigating pre-training dynamics (hyperparameter transfer, curriculum learning) on large transformer models
- Inference probing and evaluations on flow-matching policies

Research Intern

MarineAI

Jul 2025 – Sep 2025

Plymouth, UK

- Developed a reinforcement learning environment for conducting research into intelligent USV path-planning
- Utilised *pufferlib* and highly optimised C code to achieve 500K environment steps per second
- Beat established RRT/RRT* planners on both speed and adaptability

Research Intern

St Andrews University

Jun 2023 – Aug 2023

St Andrews, UK

- Developed computational models for implementing mean-field methods in open quantum systems
- Contributed to *OQuPy*, an open-source Python library for simulating open quantum systems with process tensors

PROJECTS

Trustworthy Answer Protocol (TAP) | github.com/msc-ai-mmmjo/tap

Dec 2025 – Apr 2026

- Imperial MSc group project (5 students; supervisor Dr Matthew Wicker) – post-trained OLMo-2-7B-Instruct into a multi-head “Hydra” architecture exposing calibrated uncertainty, data-poisoning security, and adversarial robustness as response-time trust signals
- Led model architecture, benchmarking, and LoRA post-training: disjoint-shard CE (security), forward-KL on AmpleGCG suffixes (robustness), uncertainty head
- Implemented Kernel Language Entropy (Nikitin et al., 2024) for free-text uncertainty using a ModernBERT-NLI surrogate over PoE response samples
- Co-developed the Product-of-Experts speculative-verification decode with custom per-head KV-cache pointer syncing, yielding a per-token security bound
- Cut adversarial flip rate from 33.4% (base 7B) to 22.8% on a 500-question GCG attack bank

Conjure-Oxide | *Rust, Python, Bash* | github.com/conjure-cp/conjure-oxide

Jan 2025 – Jun 2025

- Contributed to *conjure-oxide*, an open-source constraint programming tool written entirely in Rust
- Helped develop *tree-morph*, a type-generic Rust crate for tree transformations
- Improved algorithmic efficiency 10x by introducing a skeleton tree to remove unnecessary rule checks
- Developed wall-clock and instruction count benchmarks, integrating into CI/CD workflows via GitHub Actions

TECHNICAL SKILLS

Languages: Python, C, Rust

ML/LLM: PyTorch, HuggingFace (transformers, peft), OLMo-core, pufferlib, wandb

Developer Tools: Linux, Git, GitHub, GitHub Actions, CI/CD, Docker, pixi, Linear, Vim

AI Coding Tools: Claude Code (since Dec 2025)

PRIZES (UNIVERSITY OF ST ANDREWS)

- **Principal’s Medal:** Awarded to the top 25 final-year students across all departments
- **William David Brodie Prize:** For the highest performance in 3rd-year Physics (19.2/20.0)
- **Research Prize:** Recognising outstanding undergraduate summer research projects
- **Theoretical Physics Medal:** For exceptional performance in 3rd-year Theoretical Physics
- **Music Scholarship:** Awarded for the duration of studies